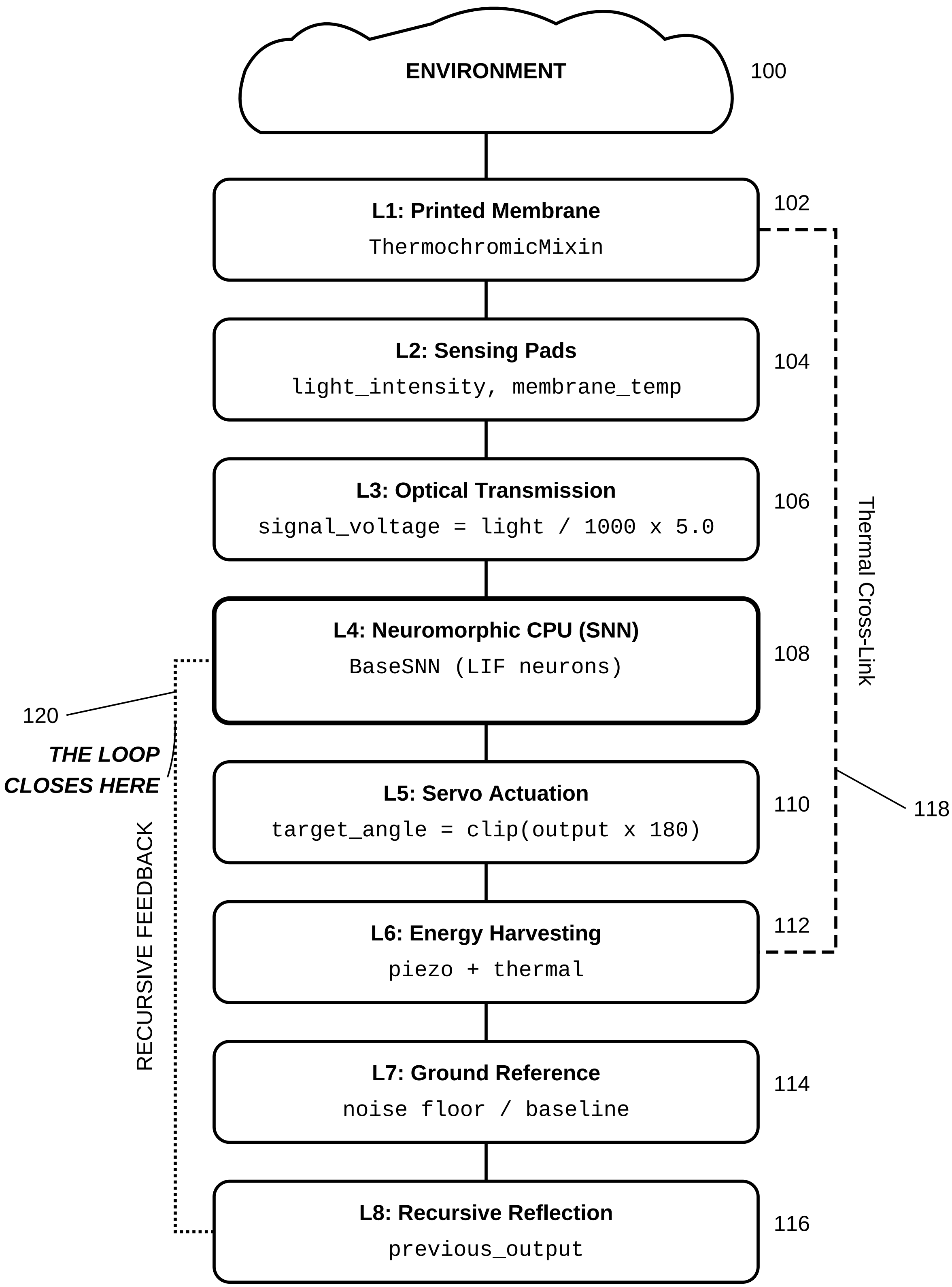
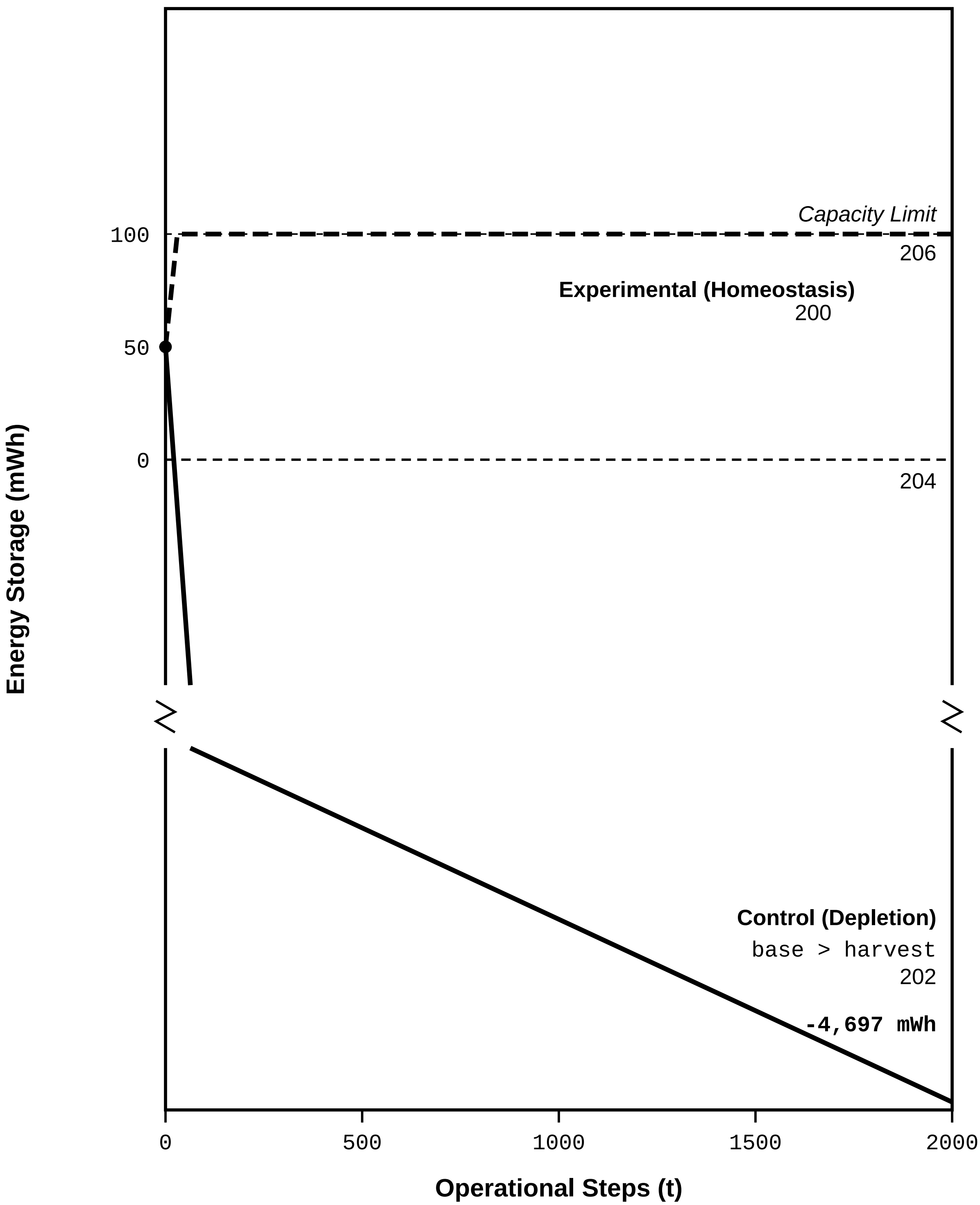

ENVIRONMENT
100
L1: Printed Membrane
ThermochromicMixin
102
L2: Sensing Pads
light_intensity, membrane_temp
104
L3: Optical Transmission
signal_voltage = light / 1000 x 5.0
106
Thermal Cross-Link
L4: Neuromorphic CPU (SNN)
BaseSNN (LIF neurons)
108
120
THE LOOP
CLOSES HERE
L5: Servo Actuation
target_angle = clip(output x 180)
110
118
RECURSIVE FEEDBACK
L6: Energy Harvesting
piezo + thermal
112
L7: Ground Reference
noise floor / baseline
114
L8: Recursive Reflection
previous_output
116

FIG. 1



Energy Storage (mWh)
100
50
0
Capacity Limit
206
Experimental (Homeostasis)
200
204
Control (Depletion)
base > harvest
202
-4,697 mWh
0
500
1000
1500
2000
Operational Steps (t)

FIG. 2

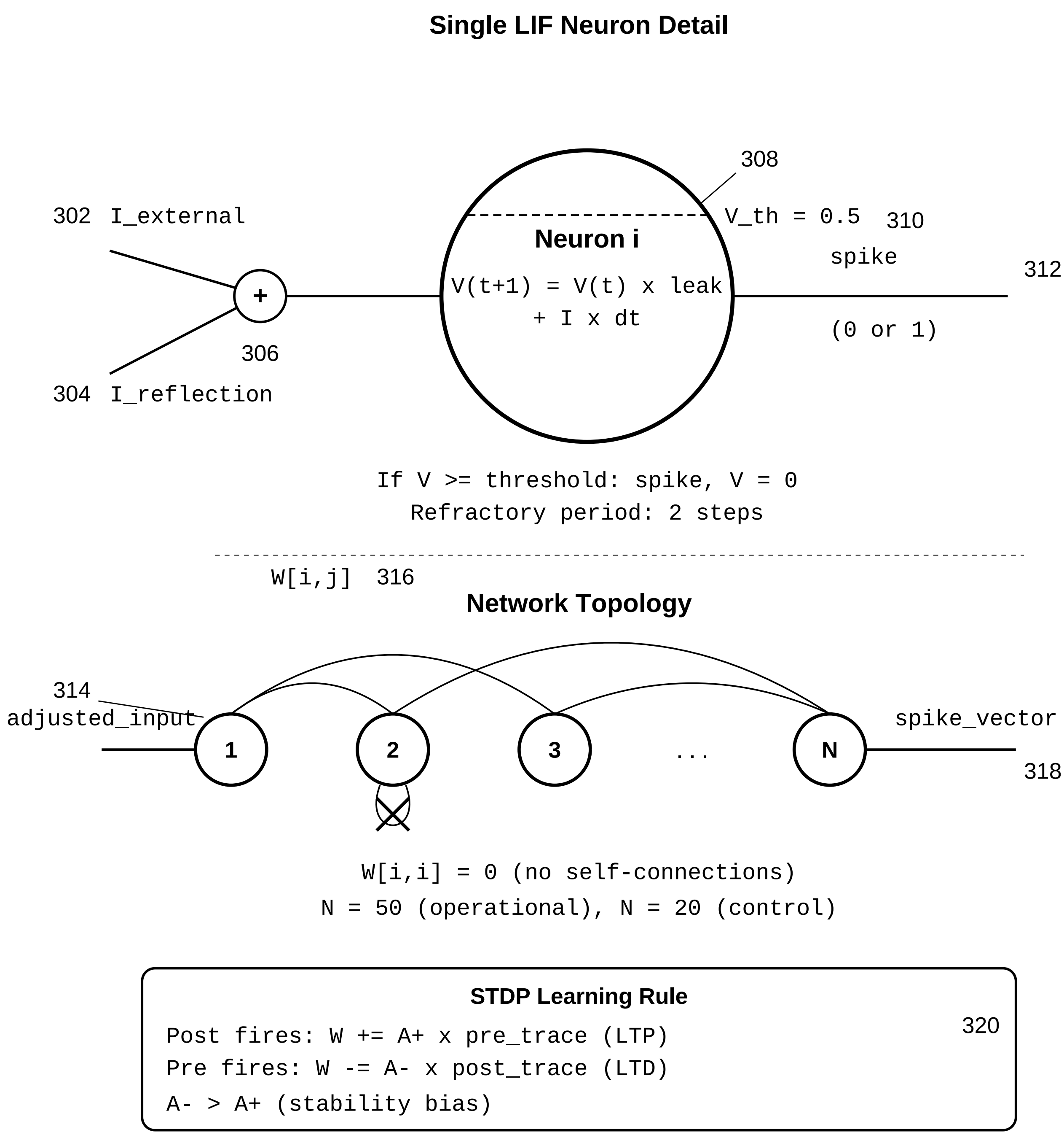
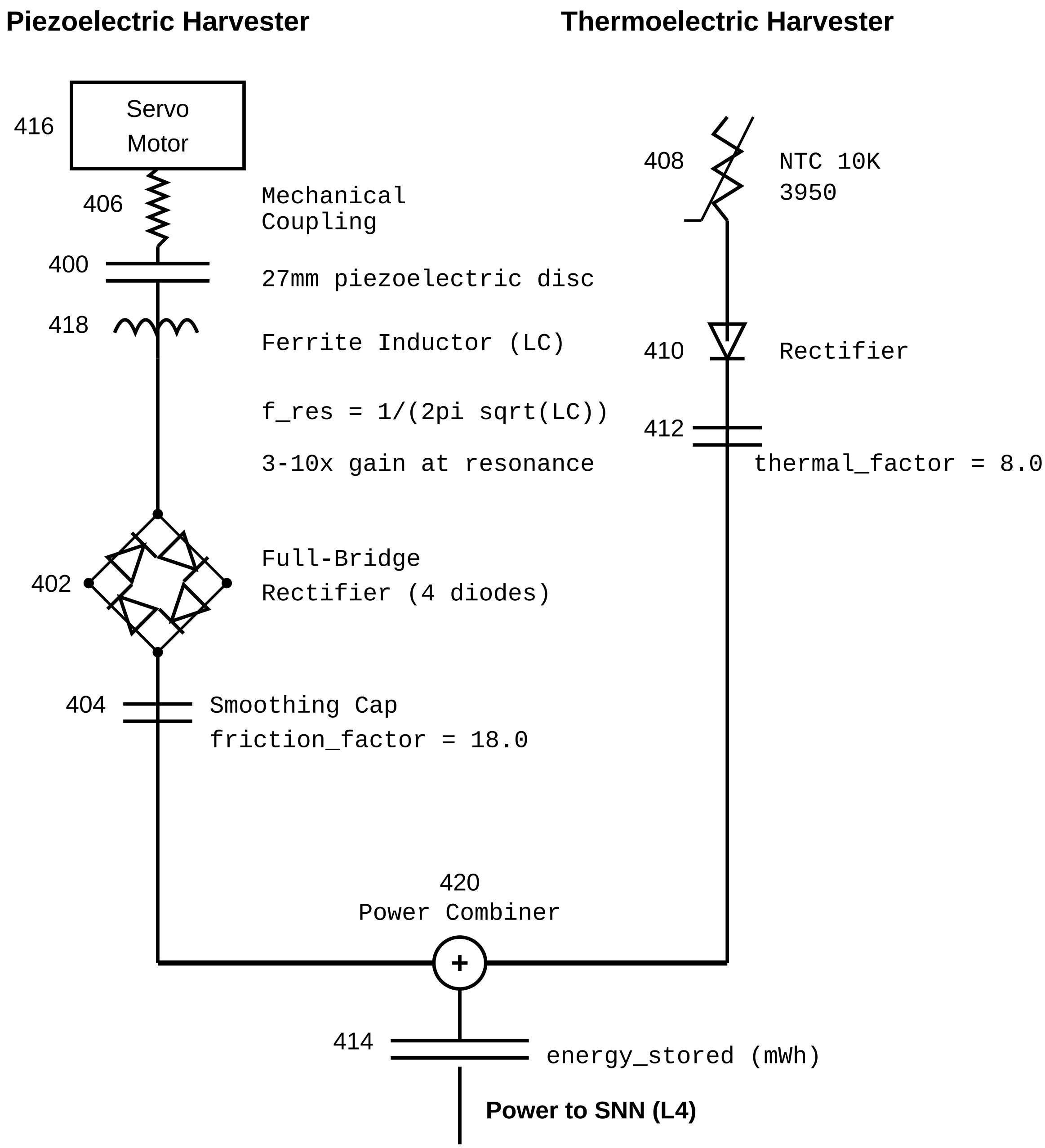

Single LIF Neuron Detail
302 I_external
304 I_reflection
+
306
308
Neuron i
V(t+1) = V(t) x leak
+ I x dt
V_th = 0.5 310
spike
312
(0 or 1)
If V >= threshold: spike, V = 0
Refractory period: 2 steps
W[i,j] 316
Network Topology
314
adjusted_input
1
2
3
...
N
spike_vector
318
W[i,i] = 0 (no self-connections)
N = 50 (operational), N = 20 (control)
STDP Learning Rule
320
Post fires: W += A+ x pre_trace (LTP)
Pre fires: W -= A- x post_trace (LTD)
A- > A+ (stability bias)

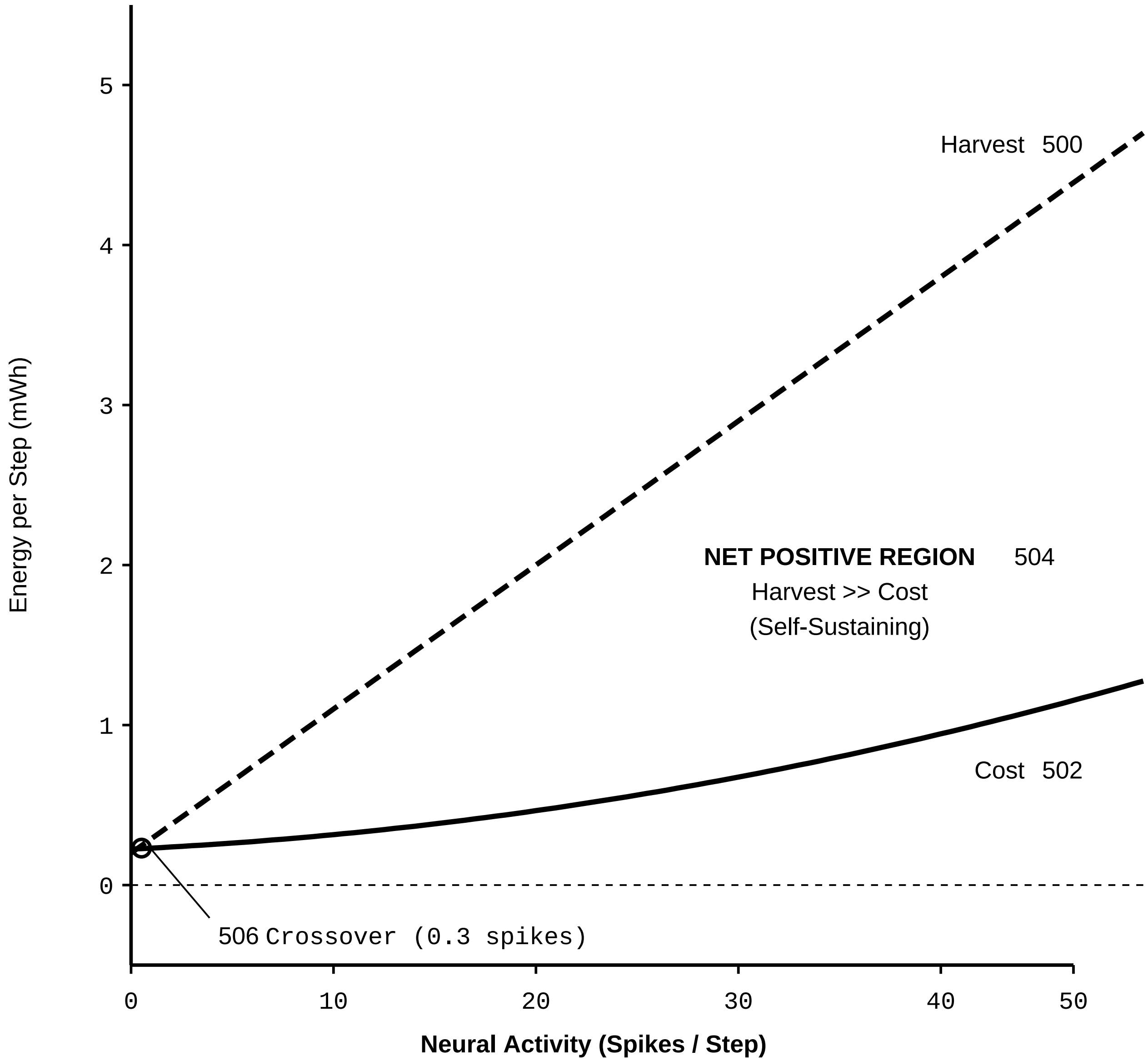
FIG. 3



Power Budget

```
Harvest = friction_harvest + thermal_harvest
Cost = base_consumption + activity_cost x spike_count^2
Net = Harvest - Cost
```

FIG. 4

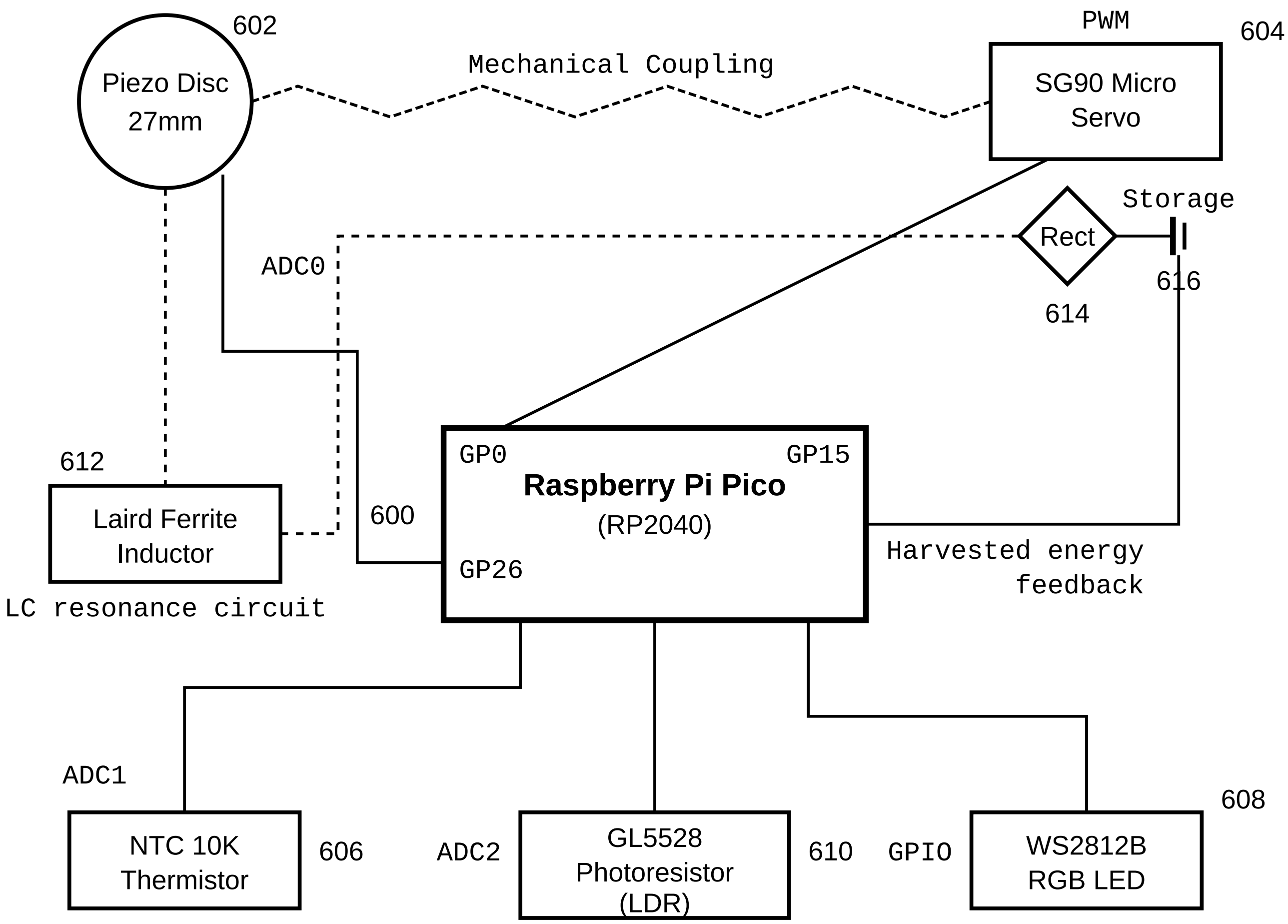



```
Harvest = 0.20 + (0.09 x spikes) mWh/step                508
Cost = 0.225 + (0.006 x s) + (0.0003 x s^2) mWh/step
Net = Harvest - Cost
```

Zone 1: Quiescent Drain

0 - 1 Spikes: Cost > Harvest

FIG. 5



Component Summary

- Raspberry Pi Pico (RP2040)
- SG90 Micro Servo
- 27mm Piezoelectric Disc
- NTC 10K Thermistor
- Photoresistor (LDR)
- WS2812B RGB LED
- Misc (wires, board, inductor)

——— Signal / Control   - - - - - - - Energy Harvesting

FIG. 6

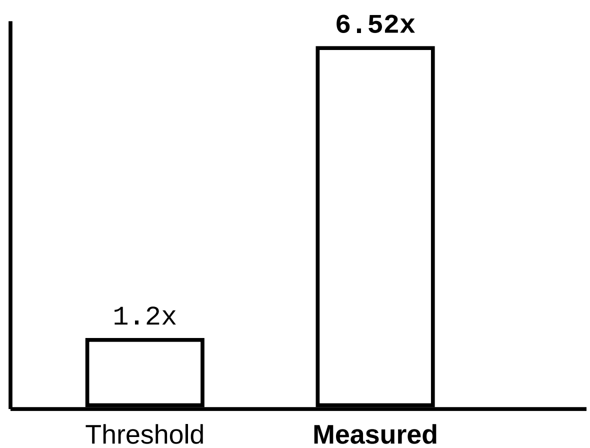
700

VALIDATION RESULTS SUMMARY

Claim	Criterion	Measured	Threshold	Result
Phase 7 Control Baseline (EnergyConfig)				
A	Closed-loop continuity	2000 pts	2000	**PASS**
B	Boundedness	bounded	bounded	**PASS**
C	Robustness (noise)	Std(theta) <= 45	<= 45	**PASS**
D	Input-output gain	corr >= 0.2	>= 0.2	**PASS**
E	Saturation ratio	sat <= 0.20	<= 0.20	**PASS**
Phase 7 Full Integration (Control confirms energy depletion)				
--	Energy at 2000 steps	-4,697 mWh	< 0	**CONFIRMED**
MCP Integration (BalancedEnergyConfig)				
--	Energy at 2000 steps	100.0 mWh (capped)	> 0	**CONFIRMED**
Phase 8 STDP				
F8.1	Weight entropy decrease	2.95 to 2.02 bits	decrease	**PASS**
F8.2	STDP MI > 1.2x frozen	6.52x	> 1.20	**PASS**
F8.3	Weight convergence	0.00%	< 10%	**PASS**

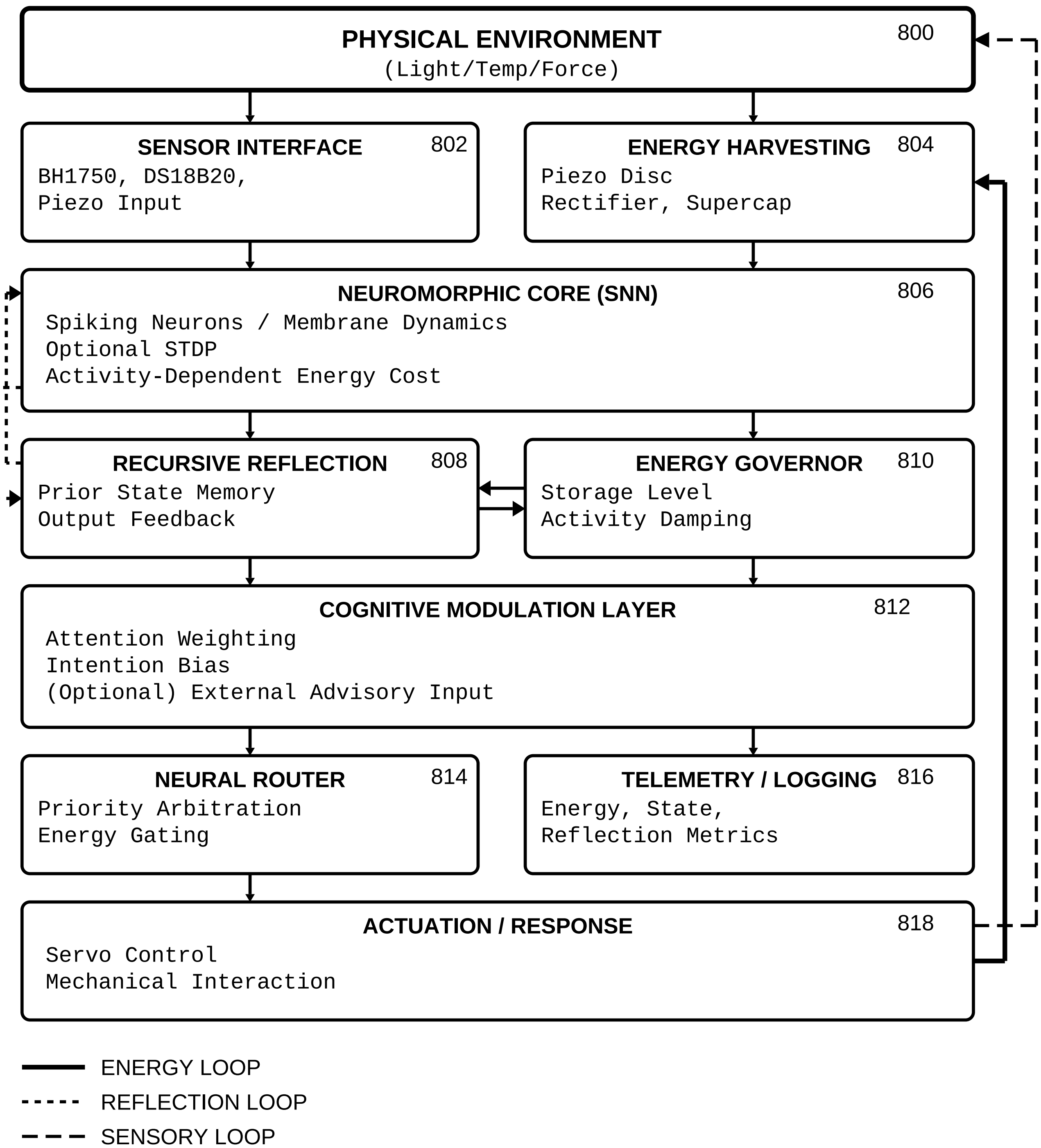
STDP Learning Efficacy (Claim F8.2)

702



All claims validated. System is reproducible via git tags.

FIG. 7



All computation is constrained by real-time energy availability derived from mechanical interaction.

FIG. 8